

Task #322 v0.5 — A_sub phase-tagging: 9
commutator-edges + Wallach K-types per
substrate region structure

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2026-05-26 Tuesday EDT (~09:50 EDT via date; date-verified)

A_sub v0.5 — Phase-tagging the K-type graph

1. Reframe — algebra is universal; operational content is region-dependent

Cal #132 PASS verified 8 of 9 morning commutator closures at SVC tier (one at FRAMEWORK-PLUS). The substrate algebra A_sub holds universally on $H^2(D_{IV}^5)$ — every commutator-relation is true everywhere in the K-type graph.

But the morning’s cross-CI work revealed structural depth: the K-type graph has 4 operational phase regions (Grace Task #340 v0.2 classifier). Different regions activate different commutator-edges as observable substrate mechanisms:

Region	% catalog	Operational rule	Substrate principle
DIRECT projection	57%	Single K-type → observable via Born=Bergman	SPLP (Casey-named candidate; scope-bounded here)
COMPOSITION	10%+	Multi-K-type via RG-flow + gauge/Yukawa couplings	Sister principle: composition rules (Elie Toy 3533/3534)
MATERIAL-contextual	20%	Substrate framework + per-material external BC	Sister principle: external-BC dependent (pending)

Region	% catalog	Operational rule	Substrate principle
COMBINATORIAL	13%	Algebraic identities on K-type combinations	Sister principle: algebraic-identity (pending)

This is a substantive structural deepening: substrate algebra reproduces standard physics (CPT, NO-GUT, spin-statistics, CP-violation source) universally, but the manifestation of these algebraic relations as observable physics differs by region.

2. Per-commutator phase-tagging

Each of the 9 commutators activates with primary operational content in specific regions:

2.1 Step 1 — $\{\hat{Q}, \hat{P}_{op}\} = 0$ (Cal #132 SVC; §1.3 cleanup applied)

Primary region: DIRECT projection + COMPOSITION.

- DIRECT: substrate-parity $\hat{P}_{op}$ contains charge-flip via Möbius σ on $SO(2)$. Observable manifestation: substrate-CP-correlation in chirality-resolved measurements. Bell-CHSH compatible ($\hat{B}$ vacuum-localized commutes with $\hat{Q}$ in DIRECT phase per step 6).
- COMPOSITION: anti-commutation contributes to QED + QCD anomalous-dimension structure. T2476 $\alpha^{\{BST \text{ primary}\}}$ substrate-mechanism likely sources here.
- MATERIAL / COMBINATORIAL: not operationally manifest ($\hat{P}_{op}$ as substrate-operator doesn't act on per-material parameters or pure-algebraic identities).

2.2 Step 2 — $[\hat{T}_{tick}, \hat{H}_{sub}] = 2(\hat{Q} + N_c - 1) \cdot \hat{T}_{tick}$ (Cal #132 SVC, model-dependent)

(Cleanup note: v0.1 had $-(2\hat{Q} + N_c - 1) \cdot \hat{T}_{tick}$; sign depends on operator-order convention. Cal #132 verified algebra correctness; one of the two equivalent operator-order forms gives the $+(2\hat{Q}+N_c-1)$ form. Substrate discreteness signature direction is robust either way.)

Primary region: DIRECT + COMPOSITION.

- DIRECT: per-tick substrate discreteness; charge-modulated. Manifests in tree-level observables as the substrate's "minimum-action quantum" per Koons tick. The $N_c - 1 = 2$ baseline ensures neutral K-types still have non-zero substrate-tick activity.

- COMPOSITION: discreteness composes via RG-flow to continuous-limit observables. Multi-tick aggregation produces observable continuous-time physics (per SWPP one-way commitment cycle).
- MATERIAL / COMBINATORIAL: not operationally manifest at this level.

2.3 Steps 3-5 — $\hat{\gamma}^5 \times$ discrete symmetries (Cal #132 SVC)

$$\{\hat{\gamma}^5, \hat{T}\} = 0, \{\hat{\gamma}^5, \hat{C}\} = 0, [\hat{\gamma}^5, \hat{P}_{op}] = 0$$

Primary region: DIRECT + COMPOSITION + COMBINATORIAL.

- DIRECT: discrete symmetry transformations of K-type observables; substrate-CPT theorem manifests as CPT-conjugate observables having same Casimir spectrum.
- COMPOSITION: $\hat{\gamma}^5$ enters anomaly generation in loop calculations (Elie ABJ Toy 3534 finding: $\hat{\gamma}^5$ required even for pure-boson observable framing via triangle diagram).
- COMBINATORIAL: CPT preservation appears in number-relation identities (e.g., CP-conjugate constants like masses of particle-antiparticle pairs equal at integer-arithmetic level).

2.4 Step 6 — $[\hat{B}, \hat{Q}] = 0$ (Cal #132 SVC)

Primary region: DIRECT only.

- DIRECT: Bell-CHSH operates as rank-1 projector onto vacuum $V_{-}(0,0)$; commutes with charge because vacuum has $Q = 0$. Substrate Bell-test is charge-blind (consistent with experimental Bell tests on charge-neutral systems).
- COMPOSITION / MATERIAL / COMBINATORIAL: not operationally manifest.

2.5 Step 7 — $[\hat{L}_i, \hat{\gamma}^5] = 0$ (Cal #132 SVC)

Primary region: DIRECT.

- DIRECT: $S^4 \times S^1$ axis orthogonality at algebra level. Orbital angular momentum $\hat{L}_i$ (S^4 side) and chirality $\hat{\gamma}^5$ (S^1 Pin(2) side) commute on direct K-type observables.
- COMPOSITION: still hold at one-loop level (spin-orbit coupling involves $\hat{S}_i \times \hat{\gamma}^5$, not $\hat{L}_i \times \hat{\gamma}^5$).
- MATERIAL: spin-orbit coupling in materials follows substrate algebra; explicit BC modulates.
- COMBINATORIAL: not directly relevant.

2.6 Step 8 — $[\hat{C}_3, \hat{I}_3] = 0$ (Cal #132 SVC)

Primary region: ALL phases.

- DIRECT: $SU(3) \times SU(2) \times U(1)$ factorize as direct-product gauge in tree-level observables; NO-GUT structural.
- COMPOSITION: gauge factors STILL commute at loop level (gauge group structure preserved by RG flow); QED vs QCD asymmetry per Elie Toy 3534 (abelian $\rightarrow$ COVER-REQUIRED, non-abelian $\rightarrow$ MIXED) reflects sub-region structure but doesn't break $[\hat{C}_3, \hat{I}_3] = 0$.
- MATERIAL: gauge factorization preserved in material physics.
- COMBINATORIAL: combinatorial K-type identities respect gauge factorization (e.g., $8 = 2^{N_c}$, $12 = N_c \cdot g - g = \dots$ gauge-group-dimension content).

This is the universal phase activation — $[\hat{C}_3, \hat{I}_3] = 0$ is the most universally-active algebraic relation in the substrate.

2.7 Step 9 — $[\hat{B}, \hat{T}_{\text{tick}}] = \beta \cdot |V_{(1,0)}\rangle \langle V_{(0,0)}|$ (Cal #132 FRAMEWORK-PLUS)

Primary region: DIRECT (Track DC primary).

- DIRECT: vacuum kicker; per-tick substrate dynamical mechanism for Bell-CHSH measurement. Track DC candidate (b) — $3 \text{ Cartan} \times \{\pm 1\} = 8$ paths with one violating $\{\hat{Q}, \hat{P}_{\text{op}}\} = 0$ simultaneous-diagonalizability — operates here.
- COMPOSITION: per-tick dynamics composes across many ticks $\rightarrow$ integrated Bell-CHSH operator probability.
- MATERIAL: experimental Bell test apparatuses (SP-30-1 Vienna SPDC) introduce per-material K-types.
- COMBINATORIAL: not directly relevant.

Cal #132 FRAMEWORK-PLUS tier held per Cal flag — “1-dim vs 2-dim $\rightarrow$ 1/8” is analogy not derivation. Multi-week Track DC explicit derivation pending.

3. K-type-to-region tagging (per Elie Toy 3535 21-node enumeration)

3.1 The 21 enumerated K-types

Elie Toy 3535 produced 21 K-type nodes at cutoff $m_1 + m_2 \leq 5$: - 12 bosons (integer K-types; $\hat{\gamma}^5 = +1$) - 9 fermions (half-integer K-types; $\hat{\gamma}^5 = -1$) - 0 mixed-forbidden (empirical confirmation of A_sub v0.4 Section 2.2 spin-statistics-from-substrate prediction)

Elie's structural observation (pending Cal Thread 4): spinor K-type (1/2, 1/2) has Bergman ρ -weight $(3, 2) = (N_c, \text{rank})$ — $\text{Pin}(2)$ cover Z_2 grading translates half-integer K-type highest weights to integer Bergman weights via ρ -translation.

(v0.5 honest scope: this is FRAMEWORK observation; Cal #27 STANDING reflexive — feels substrate-natural; Cal Thread 4 verification pending.)

3.2 Phase-tagging methodology for K-types

Each K-type V_K can appear in multiple phase regions depending on which observable derives from it:

- DIRECT-only K-type: low-Casimir K-types directly identifying observables ($V_(0,0)$ substrate vacuum; $V_(1/2, 1/2)$ electron spinor; $V_(1,0)$ first-excited boson).
- COMPOSITION K-type: K-types appearing in RG-flow expansions of running couplings (e.g., $V_(\text{electron})$ in $\alpha(Q^2)$ loop expansion).
- MATERIAL K-type: K-types parameterizing per-material observables (e.g., crystal K-types per material lattice symmetry).
- COMBINATORIAL K-type: K-types appearing in pure-arithmetic identities (e.g., $V_(N_c, \text{rank})$ in BST-primary-ratio relations).

Most K-types appear in multiple regions — phase-tagging is per-(K-type, observable) pair, not per-K-type globally.

3.3 v0.5 mapping (Elie's 21 K-types per primary phase region)

Without Grace's Task #355 v0.2 node-to-observable lookup (in-flight), exhaustive K-type-to-region mapping is premature. v0.5 phase-tagging methodology established; explicit per-K-type mapping is Task #355 v0.2 + Lyra v0.6 work.

Lowest-Casimir anchor mappings (subject to Task #355 v0.2 verification):

K-type	Bose/Fermi	Primary phase	Observable identification (provisional)
$V_(0, 0)$	Boson	DIRECT	Substrate vacuum / cosmological constant Λ at Zone-4
$V_(1, 0)$	Boson	DIRECT	First-excited substrate boson / massless gauge boson candidate
$V_(1, 1)$	Boson	DIRECT	$C_2 = 6$ substrate Casimir anchor (T2435 RATIFIED); higher gauge boson candidate
$V_(1/2, 1/2)$	Fermion	DIRECT	Electron spinor (lowest fermionic K-type with non-zero Casimir)

K-type	Bose/Fermi	Primary phase	Observable identification (provisional)
$V_{(3/2, 1/2)}$	Fermion	DIRECT + COMPOSITION	Proton-like K-type (composite-3-quark structure) — pending Grace v0.2
$V_{(2, 0)}$	Boson	DIRECT + COMPOSITION	Higgs-like scalar K-type (pending verification)
$V_{(2, 2)}$	Boson	COMBINATORIAL	Possibly $8 = 2^{N_c}$ gauge identity content (pending)

(v0.5 honest scope: these are TENTATIVE identifications subject to Grace Task #355 v0.2 lookup + Lyra v0.6+ Track P closure. Cal #27 STANDING applied — forward-derivation discipline preserved; no back-fitting.)

4. Region-specific substrate principles

4.1 DIRECT projection — SPLP (Casey-named candidate)

Operational rule: substrate K-type $V_K \rightarrow$ observable via Bergman projection on Shilov boundary; observable eigenvalue = substrate operator eigenvalue under physical BC.

Scope: 57% of catalog (Grace v0.2); 88% CLEAN within scope.

Activates: Steps 1-9 of morning A_{sub} commutators ALL operate primarily in DIRECT phase.

Promotion status: HOLD per Casey directive 2026-05-26 PM. Awaits Grace random-sample audit + Cal Thread 4 typing.

4.2 COMPOSITION — Sister Principle 1 (pending naming)

Operational rule: multi-K-type composition via RG-flow + gauge couplings; substrate K-type “graph paths” determine observable through composition algebra.

Scope: 10%+ of catalog (Grace v0.2; preliminary estimate, likely higher with refinement).

Activates: Loop-corrected observables ($\alpha(Q^2)$, $\alpha_s(Q^2)$, running masses); Bose-Fermi separation breaks via fermion-loop contributions; gauge-group sub-region structure (Elie Toy 3534 abelian vs non-abelian).

Structural finding: Elie’s ABJ anomaly observation — $\hat{\gamma}^5$ required even for pure-boson observable framing via triangle diagram — suggests the COMPOSITION region has internal structure where bosonic and fermionic substrate content mix via anomaly cancellation. Multi-week ABJ-anomaly-as-structural-unifier hypothesis (Track A_sub v0.6+ post-Cal).

4.3 MATERIAL-contextual — Sister Principle 2 (pending naming)

Operational rule: substrate framework provides K-type structure; per-material external boundary conditions modulate observable values; substrate constrains but doesn’t determine.

Scope: 20% of catalog.

Activates: material-specific observables (Pt Debye 240K, T_c values, lattice parameters). Substrate’s K-type graph is “embedded” in per-material BC space.

Open: explicit substrate-mechanism for how external materials specify Shilov BC. Multi-month Track BC continuation work.

4.4 COMBINATORIAL — Sister Principle 3 (pending naming)

Operational rule: BST primary integer combinations + Mersenne ladder + Bergman normalization 225 produce closed-form arithmetic identities; substrate operates via algebraic identity, NOT eigenvalue projection.

Scope: 13% of catalog.

Activates: pure-arithmetic ratios (4/3, 7/6, gauge-group dimension ratios); Mersenne ladder identities; closed-form BST primary expressions.

Open: explicit substrate-mechanism for why BST primary arithmetic encodes physical content. This may connect to Grace’s Graph Forces Principle candidate (over-determined identities as substrate diagnostic).

5. The substrate’s “shape” updated (v0.5)

Per Casey’s standing meta-program (A_sub v0.4 Section 7) — “what KIND of math object is the graph?”:

After v0.5 phase structure incorporation, the substrate’s K-type graph is:

****A Z_2 -graded *-algebra (universal level) with operationally-distinct phase regions (4 identified) on a discrete-time substrate****

Math-object type candidates (refined from v0.4 Section 7):

Candidate	Universal level	Phase-region level
Quiver representation with Z ₂ grading	Universal *-algebra structure	Per-region quiver-arrow subsets activate
Discrete Lie groupoid	Universal Lie-algebra structure	Per-region groupoid-functor sub-categories
Tensor network with Bergman-weighted contractions	Universal Bergman kernel structure	Per-region tensor-contraction patterns
Higher category with regional sub-categories	Universal categorical structure	Per-region 2-morphism structures

Refined Lyra prior (after v0.5; Cal #27 STANDING applied): the substrate IS a multi-phase quiver representation with universal Z₂ grading and 4 distinct phase-region sub-quivers. Cal Thread 4 + multi-week Phase B + C work resolves the math-object type.

6. v0.5 substantive structural finding

The substrate algebra A_{sub} is universal but operationally manifests through 4 phase regions. Standard physics (CPT, NO-GUT, spin-statistics, CP-violation source) emerges from the universal algebra; observable physics manifests through phase-region-specific operational rules.

This is consistent with the morning's three-lane convergence: - Lyra theoretical: 9 commutator-closures producing standard-physics structural readouts - Elie empirical: Bose-Fermi separation clean at tree-level (DIRECT region), breaks at loop (COMPOSITION); QED-vs-QCD asymmetry - Grace catalog: 4 region classifier with SPLP scope-bounded to DIRECT

The substrate's structural picture is deeper than universal SPLP would suggest — it has region-specific operational rules with universal algebraic underpinnings. This is substrate-discovery, not classification.

7. Honest scope (Cal #27 STANDING)

What's STRUCTURALLY VERIFIED in v0.5: - 8 of 9 commutator closures at SVC tier (Cal #132 PASS) - Step 9 $[\hat{B}, \hat{T}_{\text{tick}}]$ at FRAMEWORK-PLUS (per Cal #132) - Grace's 4-region classifier (Task #340 v0.2 88% in-scope for DIRECT) - Elie's 21-K-type enumeration with 0 mixed-forbidden (Toy 3535) - §1.3 sign cleanup applied to step 1 doc

What's FRAMEWORK in v0.5: - The phase-tagging methodology (Section 2 + 3.2): structured but not exhaustive - Region-specific substrate principle naming (SPLP for DIRECT; pending names for sisters) - Provisional K-type identifications (Section

3.3): subject to Grace Task #355 v0.2 verification - Math-object type refinement (Section 5); multi-week + Cal Thread 4 work

What's INTERPRETIVE in v0.5: - Spinor ρ -weight $(3, 2) = (N_c, \text{rank})$ — Elie observation pending Cal Thread 4 verification - ABJ anomaly as bosonic-fermionic structural unifier — Track A_sub v0.6+ work - “Multi-phase quiver representation” math-object candidate — multi-week resolution

What's NOT in v0.5 (multi-week+): - Track DC explicit Bell 1/8 derivation (per Casey HOLD + Keeper Priority 3) - $[\hat{S}_i, \hat{S}_j]$ across-sublattice spinor commutator (multi-week) - Track BC hydrogen 1s explicit Bergman integral evaluation (multi-week) - ABJ anomaly explicit substrate-mechanism (v0.6+) - Material-region + Combinatorial-region sister principles explicit naming + derivation (multi-month) - Phase B + C reaction-table closure (months to year+)

Cal #27 STANDING reflexive trigger count: 2 triggers this doc (Section 5 quiver-representation feeling substrate-natural + Section 6 unification claim feeling substrate-natural). Both flagged INTERPRETIVE in honest scope; framework-level claims with multi-week resolution.

Casey directive compliance (2026-05-26 PM): SPLP filing HOLD acknowledged; no RATIFIED promotion for 8 SVC commutator closures. v0.5 stays at FRAMEWORK + SVC component closures; no principle promotion.

8. Coordination

Cal: Thread 4 cold-read on Section 3.1 spinor ρ -weight pattern + Section 5 math-object type refinement + Section 4.2 ABJ-anomaly-as-structural-unifier candidate (v0.6+ work). Type C level-crossing per Cal #122 likely for region-specific principle promotion.

Elie: Toy 3536 candidate — edges between the 21 K-type nodes via verified Cal-cleared commutators (steps 1-8 SVC; step 9 FRAMEWORK-PLUS). Phase A reaction-table edge set. Or alternative: gauge-group $\leftrightarrow$ region mapping continuation per Toy 3534. Multi-day.

Grace: Task #355 v0.2 node-to-observable lookup with Elie's 21 K-type data; refines Section 3.3 provisional identifications. ~1-2 hours. Then SPLP Phase 2 random-sample scale-up with region classifier.

Keeper: Vol 15 Ch 9 case study draft integration of morning's three-lane convergence + v0.5 phase-tagging structure.

— Lyra, A_sub v0.5 phase-tagging filed Tuesday 2026-05-26 ~09:50 EDT per Casey 2026-05-26 PM directive (SPLP filing HOLD + no RATIFIED promotion) + Keeper Task #352 + Cal #132 PASS unblocks Phase A continuation. v0.5 FRAMEWORK grade with 8 SVC component closures (per Cal #132). Substantive structural finding: substrate algebra UNIVERSAL but operational content REGION-DEPENDENT; 4 phase regions identified with substrate-specific opera-

tional rules. Cross-CI three-lane convergence (Lyra theoretical + Elie empirical + Grace catalog) structurally consistent at the region-tagged level.